

AI-BASED TRAFFIC PREDICTION AND INTELLIGENT ROUTING SYSTEM

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Abstract—Urban traffic congestion is a challenge that poses an increasingly challenging challenge. Congestion in the contemporary cities, decreasing the effectiveness of commutes, growing gasoline use, and increasing green house emissions. This paper introduces a Traffic Prediction and Management System that uses AIA Long Short-Term Memory is combined with (TPMS). Congestion level classification based on a (Bi-LSTM) neural network and with smart routing interface which proposes alternative routes of expected congestion. The system is trained on a historical traffic information of various Indian cities on features that include as time of the day, weather, type of road, mean speed, and traffic density. The system represents an example of a viable and implementable smart city solution traffic management.

Index Terms—LSTM, traffic congestion prediction, smart routing, deep learning, smart city.

I. INTRODUCTION

Traffic control in the urban environment is one of the most pressing issues of the contemporary cities. The high rate of urbanisation, increasing motorisation and poor infrastructure has caused unrelenting traffic jams which have caused serious economic and environmental impacts. Urban congestion is estimated to cost economies billions of dollars each year in the lost productivity, unnecessary use of fuel, and higher vehicle emissions as per recent estimates [1].

Such classic navigation systems like Google Maps and Waze can work mainly in a *reactive* mode: these services monitor the existing situation on the road and redirect the cars. Although they work well in the short run, these systems are not able to predict future windows of congestion, thereby making suboptimal routing choices during peak-hour transitions or congestion caused by incidences in the end stages of the routes or routes to bottlenecks in an incident [2].

Sequential traffic data are best analyzed in recurrent neural networks and LSTMs in particular due to the time-dependent long-horizon modeling of data. Bidirectional LSTMs have an added advantage of processing sequences in both the forward and the reverse direction, where rich context is better captured through it, as was observed with [3]. Nonetheless, the implementation of such models in the user-facing end-to-end application should take into consideration the preprocessing pipeline or feature representation and the stability of the inference..

II. LITERATURE SURVEY

Data-driven traffic prediction The literature in data-driven traffic prediction includes statistical models, classical machine

learning, and deep learning. The most pertinent previous work is discussed in this section.

Statistical Models: Williams and Hoel [4] are among the first researchers to use Arima models to predict traffic in the short term. ARIMA is computationally lightweight and assumes that the phenomena under consideration are stationary, and it cannot readily identify non-linear behavior of traffic or capture multi variate interrelations.

Bidirectional LSTM: Cui et al. [8] have shown that Bi-LSTMs are superior to their unidirectional counterparts because they use the past and future context on a sequence window to predict the speed of bicycles in a series of traffic lanes. They are designed as a foundation of our architecture.

Prediction and Routing: A scheme that integrated LSTM-based congestion prediction with routing algorithm was proposed by Chen and Cheng [11]. Our system has a similar goal of integration but with REST-based routing API as opposed to a graph solver of their own, so they can do map-based routing on any coordinates.

Research Gap: The literature review supports the idea that (a) deep temporal models are superior to classical baselines in congestion classification, (b) Bi-LSTMs are superior to unidirectional models, and (c) end-to-end prediction-to-routing systems remain underexplored, especially in Indian urban context. All these three gaps are filled in this project.

III. PROPOSED METHODOLOGY

A. System Architecture

The pipeline of the TPMS consists of four steps, i.e., data pre-processing and features engineering, sequence building, training of Bi-LSTM models, and an inference and routing interface created using Streamlit.

B. Feature Engineering

The unprocessed data has 24 columns with each column having the temporal information, the city, name of the road, weather type, traffic volume, mean speed, reported accidents, and road closure. The target variable, which is the congestion level, has three levels that will be assessed as ordinal, which are Low, moderate and High.

The derived following engineered features are :

Cyclical Time Encoding preventing the model from treating hour 23 and hour 0 as maximally distant:

$$h_{\sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad h_{\cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right) \quad (1)$$

Rolling Mean: 3 step rolling-average of congestion per city road that provides short-term trend information.

Domain Interaction Features:

$$r_{\text{tsr}} = \frac{\text{Traffic_Density}}{\text{Avg_Speed} + 1} \quad (2)$$

$$r_{\text{se}} = \frac{\text{Avg_Speed}}{\text{Traffic_Density} + 1} \quad (3)$$

$$r_{\text{dr}} = \text{Traffic_Density} \times \text{Rush_Hour} \quad (4)$$

All categorical columns (City, Road_Type, Weather_Condition, Day_of_Week, Road_Closure, Urban_Rural) were encoded using `LabelEncoder` with encoders saved to disk for inference-time consistency. The entire dataframe — including the target column — was subsequently normalised with `MinMaxScaler`, ensuring all values lie in $[0, 1]$ before the sequence construction.

C. Sequence Construction

The data were combined based on (City, Road) and ordered by time of hour. The sequences of length $T = 30$ were extracted over-lapping, as follows in every group:

$$\mathbf{X}_i = [\mathbf{x}_i, \mathbf{x}_{i+1}, \dots, \mathbf{x}_{i+T-1}], \quad y_i = \text{Congestion_Level}_{i+T} \quad (5)$$

This provides input tensor of dimensions $(N, 30, F)$ in which F and R n features (with the name of roads dropped in each group) and N is the total length of sequences in all city-road groups.

D. Bidirectional LSTM Architecture

This model runs an individual sequence of 30 steps using two stacked layers of Bi-LSTM and finally a dense classification head:

TABLE I
BI-LSTM MODEL ARCHITECTURE

Layer	Type	Output Shape
Input	—	(30, 27)
Bi-LSTM-1	128 units, return_seq	(30, 256)
Dropout-1	$p = 0.3$	(30, 256)
Bi-LSTM-2	64 units	(128)
Dropout-2	$p = 0.3$	(128)
Dense-1	ReLU, 64 units	(64)
Dense-2 (out)	Softmax, 3 units	(3)

Loss functions are sparse categorical cross-entropy:

$$\mathcal{L} = - \sum_i y_i^* \log \hat{y}_i \quad (6)$$

where y_i^* being the actual class index and $\hat{y}_i$ being the predicted probability in that class. It was the Adam optimiser with default case learning rate of $\eta = 0.001$. The regularisation

with early stopping (patience = 8) was used to monitor the validation loss.

E. Inference Pipeline Consistency

The main implementation issue is the inference-time preprocessing is a critical issue that should precisely reflect training. The saved objects of the classifications: The saved objects of the classification encoder at inference are used on categorical input, and engineered features are computed in the same way (binding the lag and rolling features to their default values of 0). The `MinMaxScaler` is then shown a zero-filled value in place of the column which encodes the idea of `Congestion Level`, and then parameters are scaled all in the same way. Only the 27 LSTM-input columns (seq feature cols) are then fed on the model.

IV. EXPERIMENTAL RESULTS

A. Dataset Description

The data file final traffic dataset with location features CSV contains the historical traffic history of three cities in India, that include, Hyderabad, Vijayawada, and Visakhapatnam. The records are measured by one hour of traffic observation at a certain section of the road and they are full of the following fields; city, area name, road name, geographic coordinates, type of road, traffic density (0-100), average speed (km/h), weather condition, temperature ($^{\circ}\text{C}$), accident counts, availability of public transport, road blockage status, weekday, hour of the day, indicator of the rush hour, urban/rural classifications, traffic intensity and high-risk zone indicator.

The target variable is a three classes categorical variable, which is called `Congestion_Level` and whose values are: Low, Moderate, and High. Following construction of a sequence at temperature $T = 30$, the useful dataset size is about $N \approx 98,800$ samples. To avoid data leakage, an 80/20 chronological train-test split was used, and the timing is kept (`shuffle=False`).

B. Evaluation Metrics

The four weighted-average metrics are reported:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

The metrics are computed with `average='weighted'` keeping in mind for potential class imbalance.

C. Training Behaviour

The model was trained over 17 epochs when it was early stopped (best weights reinstated at epoch 9, when the validation loss had minimised to 0.4666). The accuracy on the training side risen by 0.688 of epoch 1 to 0.725 of the last epoch whereas validation accuracy leveled at 0.709, which means that the overfitting was slight yet controllable. The fact that training and validation loss are decreasing over the initial epochs supports the fact that the Bi-LSTM is acquiring temporal patterns that are meaningful of the traffic sequences.

D. Comparison with Baseline

The table below, Table II gives results in terms of the quantitative performance of the proposed Bi-LSTM model and the Logistic Regression (LR) baseline. Both the models were trained and tested with the same data splits and feature representation (sequences flattened in LR).

TABLE II
PERFORMANCE COMPARISON: BI-LSTM VS LOGISTIC REGRESSION

Method	Accuracy	Precision	Recall	F1
Logistic Regression	0.7946	0.7479	0.7946	0.7202
Bi-LSTM (Proposed)	0.8069	0.7782	0.8069	0.7599
<i>Improvement</i>	<i>+1.23%</i>	<i>+3.03%</i>	<i>+1.23%</i>	<i>+3.97%</i>

Bi-LSTM is better than the Logistic Regression in all the four metrics. The biggest improvement is on F1-score (+3.97 percentage points) and Precision (+3.03 pp) meaning that the LSTM time memory eliminates false positive alerts based on congestion, practically significant, when routing decision making, as pseudo alerts based on congestion would tend to give false alerts and thus steer drivers away needlessly. The sequential dependency structure that the Bi-LSTM is taking advantage of cannot be modelled using Logistic Regression, which uses the flattened 30-step sequence as independent features.

E. Hyperparameter Tuning

Table III and Table IV provide a summary of how validation accuracy depends on the batch size and the number of LSTM units respectively. The same 80/ 20 split with early stopping was experimented as well.

TABLE III
EFFECT OF BATCH SIZE ON VALIDATION ACCURACY

Batch Size	Val. Accuracy
16	0.699
32	0.709
64	0.701

The best relaxation accuracy was obtained using a batch size of 32 because it achieved a compromise between gradient stability and converging rate. The doubling to 64 yielded more reduced convergence but slower results with slightly less accurate results. In terms of the quantity of the LSTM units,

TABLE IV
EFFECT OF LSTM UNITS (LAYER 1) ON VALIDATION ACCURACY

LSTM Units (L1)	Val. Accuracy
64	0.697
128	0.709
256	0.705

128 in the first Bi-LSTM layer had better performance than smaller (64) and larger (256) structures; the larger structure took longer to train and a little overfitting. These empirical results led to the final configuration of Bi-LSTM(128) + Dropout(0.3) + Bi-LSTM(64) +Dropout(0.3) + Dense(64) + Softmax(3) being chosen.

The rate of dropouts was set to 0.3 according to the previous studies [8]. A sequence length of 30 was selected so that a sequence of length 30 approximately covered one day of the hourly data per road segment, both morning and evening peaks of movement..

V. DEMONSTRATION AND APPLICATION

The trained model is integrated into an interactive Streamlit web application. The interface allows users to:

- 1) Choose the mode of travel (Car or Bike) and select the start/destination points on an interactive Folium map that has its centre in Hyderabad.
- 2) Set traffic conditions (type of road, weather, traffic density, average speed, hour, etc.) using dropdowns and sliders, the possibilities of which are loaded directly out of the fitted LabelEncoder classes, and only valid training-distribution inputs are entered.
- 3) Get real-time congestion forecast (Low / Moderate / High) forecast in less than 200 ms in CPU on a model.
- 4) Be given one route (Low congestion) or as many as three different routes (Moderate/High congestion) indicated on a Folium map, in green or red to denote estimated level of congestion.

The routing backend invokes the OpenRouteService API of the type `/directions` , with the argument of `alternative_routes`. At the bottom of the page, model performance measures (Accuracy, Precision, Recall, F1) have been presented; these are presented at two decimal places

VI. CONCLUSION

The paper has proposed the Traffic Prediction and Management System based on AI incorporating a Bidirectional LSTM neural network with a live routing interface. Bi-LSTM, which was trained on 30 step hourly sequences an Indian multi-city traffic dataset with complex feature engineering, had 80.69 percent accuracy and 75.99 percent weighted F1-score on a held-out test set, outperforming the Logistic Regression base by up to 3.97 percentage points. The full end-to-end Streamlit application shows that intelligent routing and prediction of congestion can be seriously integrated into a lightweight system that can be deployed.

Limitations: The temporal memory (lag and rolling features) of the model is not entirely reproducible during inference at a single point as at a point request no historical context is available. The precision of feedback is confined to the fact that the data processed is synthetic and that there are no live sensor inputs. The topology of space of roads-networks is not modelled in the current system.

Societal and Ethical Considerations: Predictive routing systems could decrease cities congestion and its carbon footprint because of the improvement of air quality and efficiency of commutes by urban dwellers. Nonetheless, routing algorithms have the potential of causing traffic to bypass residential streets unintentionally unless alternative routes are made capacity-sensitive. Road capacity restraint should be included in the future versions. Privacy of data should also be taken into consideration in case the system is expanded to record real-time GPS tracks of users.

Future Work: Future potential directions involve in support of replacing the currently used LabelEncoder-based features with graph-structured spatial embeddings (GCN/DCRNN), the integration of real-time IoT or GPS feed, multi-step forecasting, and future research on Transformer-based networks to improve its accuracy further.

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